

The Program in Action

When you run *twentyfortyeight.py*, the output will look like this:

Twenty Forty-Eight, by Al Sweigart al@inventwithpython.com

--snip--

		2	16
	16	4	2
2		4	32
			2

Score: 80

Enter move: (WASD or Q to quit)

--snip--

How It Works

This program implements its sliding behavior using “column” data structures, represented by lists of four strings: BLANK (a single-space string), '2', '4', '8', and so on. The first value in this list represents the bottom of the column, while the last represents the top. Numbers that combine in a column always slide downward, whether the player slides the tiles up, down, left, or right. Think of it as gravity pulling the tiles in these directions. For example, Figure 79-1 shows a board with tiles sliding to the right. We’ll create four lists to represent the columns:

- ['2', '4', '8', ' ']
- [' ', ' ', ' ', '4']
- [' ', ' ', ' ', '2']
- [' ', ' ', ' ', ' ']

The `combineTilesInColumn()` function accepts one column list and returns another, with the matching numbers combined and shifted toward the bottom. The code that calls `combineTilesInColumn()` handles creating the column lists in the appropriate direction and updating the game board with the returned list.